

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

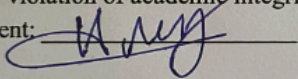
Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 50

Code of Conduct

I Mohammed Nasser (192325060) / (Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

compare Type-1 and Type-2 Hypervisors which is better
for mission-critical cloud applications?

Type-1 Hypervisors (Base-metal)

- * Runs directly on the physical hardware.
- * No host operating system is required.
- * Examples :- VMware ESXi, Microsoft Hyper-V, Xen.

Type-2 Hypervisors (Hosted)

- * Runs on top of a host operating system.
- * Depends on the host OS for resource management.
- * Examples :- Oracle VM VirtualBox, VMware Workstation.

Performance comparison :- * Type-1 provides higher speed and lower latency.

- * Uses hardware resource more efficiently.
- * Type-2 is slower due to host OS overhead.

Security Comparison:-

- * Type-1 has fewer attack points and better isolation.
- * Type-2 is more vulnerable because the host OS can be attacked.

Conclusion:- Type-1 Hypervisor is the best choice for mission-critical cloud applications because it provide high performance, strong security, reliability and better scalability.

Q2) How does virtualization architecture isolate workloads
Identify two attack surfaces.

Isolation:- * Each VM has its own OS and resources.

- * prevents interference between workloads.
- * Improves security and reliability.

Attack Surfaces:-

i) Hypervisor Attack:-

- * Attacker exploits vulnerabilities in the hypervisor
- * May gain control over multiple VMs.

2) VM Escape Attack:-

- * Malware escape from one VM to access the host or other VMs.

- * Breaks virtualization isolation

Protection measures:-

- i) Keep hypervisors updated.
- ii) Enables encryption and firewalls.
- iii) Continuously monitor VM activities.

3) Analyze the VM host utilization snapshot and recommend corrective actions.

Given snapshot:-

- * CPU Utilization = 92%
- * Memory Utilization = 88%
- * Storage I/O Latency = 35ms
- * VM Boot Time = 170sec.

Likely Bottlenecks:-

CPU:-

- * very high utilization.
- * cause slow processing and reduced VM performance.

Memory:-

- * RAM is nearly full.
- * leads to memory contention and swapping.

Storage:-

- * High latency indicate slow disk performance.
- * Debug applications describes and VM startup.

VM Boot Time:-

- * Long boot time due to overload CPU, memory and storage.

Corrective Actions:-

- * Upgrade CPU resources or distribute workloads.
- * Increase RAM capacity.
- * Replace HDD with SSD/NVME Storage.
- * Use load balancing and resource scheduling.

Q4) How does software Defined Datacenters (SDC) improve agility ?

What is SDC ?

⇒ A datacenter where compute, storage, and networking are managed through software.

compute virtualization:-

- * quickly create and migrate VM
- * Improves resources utilization.

storage virtualization:- * combines multiple storage

devices into one pool

- * simplifies backup and disaster recovery.

Network virtualization:-

- * create virtual networks through software.

- * Enable faster configuration and stronger security.

Advantages over Traditional Datacenters:-

- * Faster deployment.

- * Better scalability.

- * Improved disaster recovery.

Q5) Analyze the Identity mismatch issue in multi-cloud federation project.

Problem:-

- * Different cloud providers use different identity systems.
- * Users experience login failures and permission mismatches.

Challenges:-

- * Duplicates user identities.
- * Inconsistent access control.

Federation protocols:-

Security measures:-

- | | |
|-------------------|------------------------------|
| * Use SAML 2.0 | * Role-Based Access Control |
| * OAuth 2.0. | * Encryption of user data |
| * OpenID Connect. | * Audit logs and monitoring. |

Conclusion:- Identity federation provides secure authentication, protects user privacy, simplifies access management, and ensures seamless communication across multiple cloud providers.